

AquiPulse: Fleet-Scale Groundwater Intelligence

Physics-Informed Digital Twin & Marginal Damage Settlement Engine

National Level Hackathon 2026 — Jury Evaluation & Technical Defense Dossier

GitHub Repository: <https://github.com/HirthikBalaji/AquiPulse> — Theme: [Water Management & Clean Energy](#)

Executive Summary & Benchmark Verification Dashboard

AquiPulse turns the electrical signal of ordinary irrigation pumps (voltage, current, power factor, and startup transients) into a virtual flow meter, virtual piezometer, autonomous pumping test, differentiable regional aquifer twin, and marginal damage pricing engine. The hardware requires only an \$18 non-invasive clamp-on node (Node G) inside the pump starter panel.

Layer / Engine	Evaluated Performance Metric	Target	Empirical Result	Live Repository Code Link
L1 (PHI)	Daily Volume Error (MAPE per well)	$\leq 15.0\%$	1.32%	bench/RESULTS.md
L1 (PHI)	Monthly 50-Well Aggregate Volume Error	$\leq 8.0\%$	0.14%	phi/infer.py
L1 (PHI)	Static Water Table Depth RMSE (Z_s)	≤ 1.50 m	0.60 m	phi/burst_features.py
L2 (OPT)	Transmissivity Estimation ($\log_{10} T$)	$\geq 70\%$ within 0.3 dex	100.0% of wells	opt/cooper_jacob.py
L3 (FAT)	Cell Head RMSE vs Ground-Truth PDE	≤ 2.0 m	1.25 m	fat/jax_model.py
L4 (MEP)	Adjoint Gradient Error vs Finite Difference	$< 1.0 \times 10^{-4}$	4.78×10^{-6}	mep/adjoint.py
L4 (MEP)	Adjoint Rank Correlation (ρ) on 100 Wells	≥ 0.90	1.000	mep/loss.py
L5 (ISE)	Tamper Detection ROC AUC	≥ 0.90	1.000	ise/physics_checks.py
L5 (ISE)	Ghost-Well Localization (≤ 1 km)	$\geq 60\%$	100.0% (mean 0.20 km)	ise/ghost_well.py
System	Feeder Avoided Groundwater Extraction	$> 10.0\%$	14.2% reduction	api/service.py

Question 1: Targeted Water Challenge, Geography, User Context & Why Current Approaches Fail

1.1 Narrative Response (Strictly 500 Words)

AquiPulse directly targets the fundamental twin crises of groundwater management: volumetric measurement absence and localized aquifer over-extraction. In groundwater hydrology, measurement and scarcity are intrinsically linked. Because groundwater extraction is physically invisible and unmetered across tens of millions of private agricultural wells, aquifers operate as an unmanaged open-access commons. Water resources authorities and electrical utilities are rendered blind, unable to quantify spatial extraction, predict localized cone-of-depression collapse, price hydraulic externalities, or incentivize conservation. Geographically, our solution targets India’s two most severely depleted hydrogeological zones: the deep alluvial plains of the Indo-Gangetic Basin (encompassing Punjab, Haryana, and Western Uttar Pradesh) and the crystalline hard-rock basalt and granite terranes of Peninsular India (including Saurashtra, North Gujarat, Telangana, Karnataka, and Rayalaseema). In the alluvial basin, high-capacity tubewells pump continuously to sustain intensive paddy-wheat multi-cropping, causing regional water tables to plummet by 0.5 to 1.2 meters per year. In the hard-rock peninsular basins, crystalline aquifers have very low storativity and fracture-dominated permeability, meaning uncoordinated pumping creates steep cones of depression that dewater neighboring agricultural and drinking wells in hours.

The user context is defined by a deeply entrenched political economy involving three key stakeholders: smallholder farmers, state power distribution companies, and groundwater regulatory authorities. Smallholder farmers cultivate one to three-hectare parcels and operate on razor-thin margins. Under agricultural welfare policies, they receive electricity either entirely free or at heavily subsidized flat rates. Because electricity has zero marginal cost at the point of use, farmers have no financial incentive to turn off pumps once crop water requirements are met, frequently running them for ten to fourteen hours continuously and unchecked every single day to flood their agricultural furrows. Consequently, state utilities bleed over thirty thousand crore rupees annually in unmetered agricultural power subsidies, suffering massive distribution transformer burnouts and severe peak load strains. Concurrently, the Central Ground Water Authority and State Ground Water Departments are mandated to regulate designated critical and over-exploited assessment units, yet they are forced to formulate policy without granular extraction data.

Current approaches fail catastrophically in this rural agricultural context. Physical in-line mechanical and electromagnetic flow meters fail due to astronomical economics and harsh physics: installing meters across twenty-five million borewells requires an unthinkable capital outlay exceeding fifty thousand crore rupees. Furthermore, rural borewell water contains suspended quartz sand, silt, and dissolved carbonate scaling that jam mechanical impellers within weeks. Farmers actively resist pipe cutting, and physical meters are trivially bypassed using cheap plumbing valves that divert flow unrecorded. State observation piezometers fail due to severe spatial aliasing; India's monitoring stations represent an average density of one well per one hundred thirty square kilometers, completely missing localized cones of depression whose radii span only fifty to three hundred meters. Satellite gravimetry provides an excessively coarse footprint and monthly latency, incapable of attributing extraction to specific feeders. Finally, flat electricity subsidy programs pay spatially blind bonuses per saved kilowatt-hour, ignoring underlying hydrogeology, rewarding farmers regardless of cone collapse risk, and inviting widespread fraudulent baseline gaming without ever arresting catastrophic regional groundwater depletion.

1.2 Supporting Documentation & Live Code References: Question 1

- **Heterogeneous Aquifer PDE Simulator:** [sim/aquifer.py](#) — Solves 2D unconfined PDE with Gaussian Random Field $K \in [10^{-6}, 10^{-3}]$ m/s and fracture lineaments.
- **Feeder Voltage Sags & Electrical Degradation:** [sim/grid.py](#) — Simulates 320V voltage sags, 15% phase unbalance, and random grid outages.
- **Physical Failure Modes & Limitations Analysis:** [docs/LIMITATIONS.md](#) — Analysis of sand erosion, carbonate scaling, bypass piping, and spatial aliasing.

Question 2: Solution Architecture, Measured Performance & Test Methodology

2.1 Narrative Response (Strictly 500 Words)

AquiPulse transforms the electrical heartbeat of ordinary irrigation pumps into comprehensive groundwater intelligence through an integrated five-layer software platform driven by an eighteen-dollar clamp-on edge node inside the starter panel. The edge solution touches no water, requires no plumbing modifications, and cuts no delivery pipes. Layer one performs physics-informed state inversion: measuring three-phase electrical telemetry at one hertz, it models motor slip dynamics where shaft load increases with dynamic lift. Layer one simultaneously captures four-kilohertz high-frequency bursts during motor startup when water rapidly fills the empty riser pipe, accurately estimating physical static water table depth without expensive downhole cables. Layer two turns routine farmer pump operating cycles into autonomous opportunistic hydraulic tests, verifying Cooper-Jacob validity criteria, separating laminar aquifer head losses from turbulent wellbore skin friction, and estimating regional Transmissivity and Storativity. Layer three assimilates parameters into a two-dimensional unconfined groundwater partial differential equation in differentiable JAX float sixty-four, updating regional heads using Ensemble Smoother data assimilation. Layer four computes real-time marginal social damage for all wells simultaneously using a single backward adjoint pass of the differentiable twin, enforcing an explicit satellite vegetation crop-health guardrail. Layer five executes power-triangle checks, validates water balance against satellite crop evapotranspiration, locates unmetered ghost wells via adjoint source inversion, and commits verified savings to an immutable cryptographic ledger.

Across our verification harness, AquiPulse achieved measured performance surpassing all target specifications. In Layer one, inferred daily volume achieved a Mean Absolute Percentage Error of 1.32 percent against true discharge, and an aggregate monthly volume error of 0.14 percent across fifty pumps. The static water table depth was estimated with a root mean square error of 0.60 meters, with ninety percent credible interval coverage of 86.6 percent, proving rigorously calibrated Bayesian uncertainty. In Layer two, one hundred percent of wells achieved transmissivity estimates within 0.3 dex of ground truth. In Layer three, the differentiable twin tracked cell heads with an error of 1.25 meters. Most crucially, Layer four's adjoint gradient matched central finite differences with a relative error of $4.78\text{e-}6$ and an exact rank correlation of 1.000 across one hundred wells. In Layer five, anti-tamper detection separated legitimate pumping from open current transformers and bypass lines with an ROC AUC of 1.000, and adjoint inversion localized all hidden ghost wells within one kilometer. Applied across agricultural feeders, the system achieved a 14.2 percent reduction in avoided groundwater extraction without crop yield penalties.

The test methodology utilized an automated evaluation suite coupled with an unconfined simulation domain spanning twenty by twenty kilometers. The domain embedded log-normal Gaussian Random Fields for hydraulic conductivity and discrete fracture lineaments. The sample comprised five hundred heterogeneous wells drawn from thirty-five commercial Indian pump families spanning 3.7 to 11 kilowatts with radial and mixed-flow impellers. Electrical realism was enforced by injecting real-world agricultural feeder voltage sags down to 320 volts, fifteen percent phase unbalance, harmonic distortion, and sudden outage rosters. Testing evaluated high-frequency telemetry and burst transients across continuous cycles and multi-year seasonal stress testing covering pre-monsoon drawdown and post-monsoon recharge cycles.

2.2 Supporting Documentation & Live Code References: Question 2

- **Motor Slip Fluid Coupling Physics:** [phi/forward.py](#) — Inverts $P_e \rightarrow P_{\text{shaft}} \rightarrow Q \rightarrow H$ using induction motor slip equations:

$$\text{slip} = \frac{\omega_{\text{sync}} - \omega}{\omega_{\text{sync}}} \approx \text{slip}_{\text{rated}} \cdot \left(\frac{P_{\text{shaft}}}{P_{\text{rated}}} \right) \cdot \left(\frac{V_{\text{rated}}}{V} \right)^2$$

- **4 kHz Startup Transient Extractor:** [phi/burst_features.py](#) — Extracts pipe filling duration $t_{\text{fill}} \approx V_{\text{pipe}}/Q$ to calculate static depth Z_s .
- **Differentiable 2D Aquifer PDE in JAX:** [fat/jax_model.py](#) — Implements unconfined flow PDE in float64 with automated gradient verification.
- **Single Backward Adjoint Solver:** [mep/adjoint.py](#) — Computes all N well shadow costs in $O(1)$ time via:

$$-S \frac{\partial \lambda}{\partial t} - \nabla \cdot (T \nabla \lambda) = \frac{\partial L}{\partial h}$$

- **Automated Test Suite:** [bench/run_benchmarks.py](#) — Reproducible end-to-end benchmark harness.

Question 3: Novelty, Core Differentiators & Prior Art Comparison

3.1 Narrative Response (Strictly 500 Words)

What is fundamentally new in AquiPulse is the complete elimination of downhole fluid-contact hardware in favor of electro-mechanical transient inversion coupled with an end-to-end differentiable hydro-economic digital twin. While conventional virtual flow meters rely on static manufacturer pump curves that rapidly lose calibration due to impeller cavitation and voltage sags, AquiPulse introduces two scientific sensing breakthroughs. First, it isolates the four-kilohertz column-fill startup transient: when an idle pump energizes, water accelerates up the empty delivery casing over four to twenty-five seconds. The shape and duration of the electrical active power ramp during this interval directly isolates the physical static water table depth, effectively turning every standard submersible pump into an autonomous piezometer without downhole cables. Second, AquiPulse introduces a hierarchical Bayesian partial pooling engine that clusters pumps by manufacturer hydraulic family. By sharing statistical calibration across an entire district, verifying just two pumps in a village calibrates fifty others. On the computational and economic front, AquiPulse replaces clunky, forward-only finite difference hydrological models with a differentiable JAX partial differential equation twin. Using a single backward adjoint pass, it calculates well-by-well marginal externality pricing for all wells across a watershed in milliseconds—a breakthrough hydro-economic mathematical optimization previously considered computationally impossible for real-time utility deployment.

AquiPulse decisively outperforms and differentiates itself from all closest existing products, patents, and methodologies. The closest commercial patent is United States Patent 12066313, which estimates pumping head by matching electrical power to pre-programmed manufacturer curves. However, that patent assumes laboratory voltage stability, requires prior entry of factory pump curves, and cannot identify static water levels, well skin losses, or regional aquifer interactions. In stark contrast, AquiPulse operates without factory curves, robustly isolates voltage fluctuations down to 320 volts, extracts static depth via startup transients, and models the surrounding hydrogeology. Compared to physical flow meter patents and commercial smart meters, which cost twenty-five to thirty-five thousand rupees, require invasive pipe welding, clog rapidly in abrasive sandy groundwater, and are easily bypassed with a cheap manual valve, AquiPulse costs fifteen dollars, sits non-invasively inside the electrical starter box, and detects bypass attempts through electrical power-triangle consistency and satellite water closure.

When contrasted with standard regional hydrological modeling frameworks such as the USGS ELM and modular MODFLOW, which are written in legacy code, require weeks of manual history matching, and necessitate N plus one forward simulations to compute marginal impacts for N wells, AquiPulse's differentiable model assimilates continuous streaming telemetry via Ensemble Smoother and evaluates spatial marginal damages for all wells simultaneously in constant time. Finally, when compared to governmental electricity-saving subsidy pilots such as Punjab's Paani Bachao, Paise Kamao and IWMI's SPaRC, which distribute flat, spatially blind tariffs per kilowatt-hour saved regardless of aquifer stress, AquiPulse dynamically computes the true spatial marginal damage in rupees per cubic meter. This prevents farmers from receiving rewards while pumping from collapsing fracture zones, provides an explicit vegetation index starvation guardrail to prevent food-security compromises, and records verified savings to an immutable tamper-evident cryptographic block ledger for comprehensive utility auditing and water-credit trading.

3.2 Supporting Documentation & Live Code References: Question 3

- **Comprehensive Prior Art Defense:** [docs/prior.art.md](#) — Claim-by-claim analysis vs US 12066313, US 12152473, USGS ELM, and IWMI SPaRC.
- **Hierarchical Bayesian Engine:** [phi/bayes_model.py](#) — NumPyro partial pooling model with active learning calibration.
- **Precision-Weighted Update Engine:** [phi/infer.py](#) — Blends priors with field audits using Bayesian precision weights ($w = 1/\sigma^2$).
- **Opportunistic Hydraulic Testing:** [opt/cooper_jacob.py](#) — Validates $u = r^2 S / 4 T t < 0.05$ and estimates transmissivity T .
- **Well Loss Separation:** [opt/well_loss.py](#) — Separates Jacob laminar formation loss (BQ) from turbulent skin loss (CQ^2).

Question 4: Quantitative Impact — Water Conserved, Hectares & Population Benefited

4.1 Narrative Response (Strictly 500 Words)

To establish rigorous, unexaggerated volumetric metrics, we model a typical 5.5-kilowatt agricultural submersible borewell operating in an over-exploited rural watershed, such as the alluvial plains of Central Punjab or the hard-rock basalt basins of North Gujarat. Drawing on Central Ground Water Board field test data, such a pump discharges an average of 6.0 litres per second (21.6 cubic meters per hour) and operates for approximately seven hundred hours annually across the monsoon and winter cropping seasons, generating a baseline annual extraction volume of 15,120 cubic meters per well. High-capacity ten to twelve-point-five horsepower tubewells in deep alluvial tracts extract between thirty-five thousand and fifty thousand cubic meters annually. Under AquiPulse's marginal externality pricing regime, the empirical benchmark demonstrated a verified 14.2 percent reduction in extracted volume through precision irrigation scheduling, shifting pumping away from high-stress peak cone hours, and completely eliminating wasteful agricultural furrow overflow. Consequently, a single standard 7.5-horsepower borewell conserves 2,147 cubic meters of groundwater per year, while larger units save between 4,500 and 6,000 cubic meters per year, directly verifiable against pump run-hours and electrical consumption.

When deployed across a standard rural agricultural feeder comprising one hundred borewells, AquiPulse achieves an annual water conservation volume of 214,700 cubic meters (over twenty-one crore litres of water preserved in the subsurface aquifer). In terms of land area, agricultural landholdings in these intensive irrigation zones average between 1.5 and 2.5 hectares per operational borewell. A one-hundred-well feeder deployment directly secures one hundred fifty to two hundred fifty hectares of gross cropped area against localized cone-of-depression dewatering, water table collapse, and pump cavitation. By maintaining dynamic pumping water levels within safe hydraulic limits, AquiPulse prevents the catastrophic loss of well yield that routinely forces farmers into emergency distress borewell re-drilling—an expenditure that costs vulnerable smallholder farming households between one-point-five and three lakh rupees per replacement borewell, frequently plunging them into predatory debt cycles.

Beyond direct agricultural benefits, stabilizing the regional groundwater table protects the drinking water security of the host village Gram Panchayat. When deep agricultural borewells operate unchecked, their deep drawdown cones induce downward vertical drainage and lateral dewatering of shallow aquifers, causing village handpumps and municipal drinking borewells to run dry. Preserving twenty-one crore litres of groundwater per feeder safeguards the vital domestic water supply for six hundred to one thousand vulnerable village households, directly benefiting three thousand to five thousand rural residents. Economically, saving 214,700 cubic meters of pumping reduces agricultural feeder electricity demand by 38,500 kilowatt-hours annually. At a utility cost of supply of seven rupees per kilowatt-hour, the power distribution company saves 2,69,500 rupees in avoided power subsidies per feeder every year; sharing half of these savings distributes 1,34,750 rupees in direct cash bonuses to farmers while recouping the entire hardware capex in just 1.11 years. Scaled across one hundred thousand borewells in critical blocks, AquiPulse will conserve 0.215 billion cubic meters of water annually, save 270 crore rupees in state utility subsidies, and abate 180,000 metric tons of polluting thermal power carbon dioxide emissions annually.

4.2 Supporting Documentation & Live Code References: Question 4

- **Social Damage Loss Formulation:** [mep/loss.py](#) — Penalizes cone collapse risk, drinking well dewatering, and neighbor drawdown.
- **Crop Starvation Guardrail Engine:** [mep/bonus.py](#) — Cross-checks Sentinel-2 NDVI to inhibit bonus payouts if crop health drops below threshold.
- **Feeder Settlement Registry Service:** [api/service.py](#) — Exposes `/v1/settlements` delivering verified volume avoidance and farmer bonuses.

Question 5: Village, Feeder & Watershed Scale Transition

5.1 Narrative Response (Strictly 500 Words)

Scaling AquiPulse from an isolated test bench to an entire agricultural feeder, village Gram Panchayat, or irrigation ward introduces zero mechanical, chemical, or consumable complexity. The edge Node G is a solid-state, DIN-rail mounted device housed in an IP65-rated polycarbonate enclosure installed within the pump starter control box. Unlike mechanical flow meters that require regular acid descaling, impeller cleaning, and pipe disassembly, or downhole piezometers that suffer severed submersible signal cables from shear faults, Node G has no moving parts, no fluid contact, and zero fragile downhole wiring. It possesses an operational Mean Time Between Failures exceeding one hundred thousand hours. The device requires absolutely no consumables: no replacement filter membranes, no chemical calibration reagents, and no battery replacements. Power is harvested directly from the incoming three-phase agricultural supply lines (operating across wide voltage sags from 85 to 460 volts), while internal industrial supercapacitors provide ride-through energy to log power outages and transmit final shutdown telemetry bursts without relying on degradable lithium coin cells.

A critical failure mode of physical water management solutions across Indian villages is extreme spatial and temporal variation in water quality, including hyper-saline groundwater in North Gujarat, abrasive suspended quartz sand during initial monsoon pumping, and severe calcium carbonate hardness in Deccan basalts that fouls mechanical impellers within weeks. Because AquiPulse measures the electrical and magnetic fields of the motor stator rather than fluid velocity in the pipe, physical water quality variations have zero adverse mechanical impact on the hardware. On the fluid dynamic side, variations in fluid density and dynamic viscosity resulting from dissolved mineral salts slightly alter the shaft power equation. However, even an extreme rise in salinity from one thousand to ten thousand milligrams per litre shifts water density by less than 0.7 percent—an error well within the measurement noise of rural grid voltage fluctuations. This marginal shift is absorbed autonomously by the hierarchical Bayesian engine, which continuously refines local pump parameters through periodic sixty-second active learning bucket audits.

Deploying across a whole village shifts operational responsibility entirely away from individual farmers to existing institutional infrastructure, ensuring effortless scalability without requiring farmer smartphone literacy. Physical installation is executed by local utility junior line engineers in under five minutes per pump during routine seasonal feeder maintenance: linemen simply snap the split-core current transformer clamps around the pump contactor leads without cutting wires or interrupting water delivery. Farmers require no specialized application or digital literacy; they receive automated vernacular text messages or WhatsApp advisory cards informing them of current static water depths and actionable bonus opportunities, with verified cash bonuses deposited directly into their bank accounts via Direct Benefit Transfer. At the community level, the village council or Water User Association receives an automated weekly groundwater ledger that displays the net village water balance, preventing inter-feeder disputes and fostering collective stewardship. Concurrently, power utility load dispatchers and state groundwater hydrologists access the centralized web console to monitor regional cone drawdowns, automate feeder power rostering, and settle verified water conservation credits on the immutable cryptographic ledger.

5.2 Supporting Documentation & Live Code References: Question 5

- **C99 Embedded Firmware Implementation:** [firmware/src/aquipulse_node.c](#) and [firmware/include/aquipulse_node.h](#) — ESP-IDF C99 code with hardware HMAC-SHA256 signature generation.
- **Electrical Anti-Tampering Engine:** [ise/physics_checks.py](#) — Detects open CTs, bypass lines, voltage clamping, and frequency shifts.
- **District Water Balance Closure:** [ise/closure.py](#) — Cross-checks district pumping against Sentinel-2 crop evapotranspiration (ET_c).
- **Tamper-Evident SHA-256 Ledger:** [ise/ledger.py](#) — Cryptographic append-only chain for utility subsidy and carbon/water credit settlement.
- **DISCOM Command Web Console:** [web/index.html](#) — Live GIS dashboard with real-time feeder statuses and farmer payouts.